KERALA SSLC | MATHEMATICS

TRIGONOMETRY

PYQ • 2021–2026

1. 2026 **4 MARKS**

In triangle ABC, AB = 5 centimetres, BC = 8 centimetres and ∠B = 140°. Calculate the area of triangle ABC.

2. 2026 **5 MARKS**

From the top of a mobile tower, two steel wires are stretched to either side and attached to the ground, making angles 70° and 48°. The distance between the feet of the wires is 40 metres.

(i) Draw a rough sketch showing these details. **(1)**

(ii) What is the height of the tower? **(4)**

OR

3. 2026 **5 MARKS**

In a triangle, one side is 10 centimetres and the angles on that side are 60° and 45°. Calculate the area of the triangle.

4. 2025 **4 MARKS**

In triangle ABC, AB = 8 centimetres, BC = 10 centimetres and ∠B = 50°.

(a) Find the perpendicular distance from the vertex A to BC.

(b) Find the area of triangle ABC.

(sin 50° = 0.77, cos 50° = 0.64)

5. 2025 **5 MARKS**

A boy standing away from the foot of a building sees the top of the building at an angle of elevation 35°. Stepping 25 metres forward he sees the top of the building at an angle of elevation 70°. Height of the boy is 1.6 metre.

(a) Draw a rough figure based on the given details.

(b) What is the distance between the boy and the building when he sees the top at an elevation of 70°?

(c) Find the height of the building.

(sin 35° = 0.57, cos 35° = 0.81, tan 35° = 0.70, sin 70° = 0.94, cos 70° = 0.34, tan 70° = 2.7)

6. 2025 **4 MARKS**

In trigonometry the reciprocals of sin and cos are called cosecant and secant. The reciprocal of tan is called cotangent. They are shortened as cosec, sec and cot.

Hence, cosec x = 1/sin x, sec x = 1/cos x, cot x = 1/tan x

Answer the following questions based on the above details.

(a) sin x × cosec x = _____

(b) cosec 60° = _____

(c) cot 45° = _____

(d) What is sec 60° – cosec 30°?

(1, 0, 1/2, √3/2)

7. 2024 **4 MARKS**

The diagonal of a rectangle is 9 centimetres and it makes an angle 49° with one side. Find the length of the sides of the rectangle.

(sin 49° = 0.75, cos 49° = 0.66)

8. 2024 **5 MARKS**

From a point on the ground at a distance of 100 metres away from a tower, the top of the tower is seen at an angle of elevation 45°. From the top of the tower, a car is seen on the opposite side of the tower at an angle of depression 25°.

(a) Draw a rough figure showing the details given in the question.

(b) Find the height of the tower.

(c) What is the distance of the car from the tower?

(sin 65° = 0.91, cos 65° = 0.42, tan 65° = 2.14)

9. 2023 **4 MARKS**

The sides of a right triangle are 9 centimetres, 12 centimetres and 15 centimetres.

(a) Find the area of the triangle.

(b) Calculate the inradius of the triangle.

10. 2023 **5 MARKS**

A child sees the top of a telephone tower at an elevation of 80°. Stepping 20 metres back, he sees it at an elevation of 40°.

(a) Draw a rough figure.

(b) Calculate the height of the tower.

(sin 40° = 0.64, cos 40° = 0.77, tan 40° = 0.84, sin 80° = 0.98, cos 80° = 0.17, tan 80° = 5.7)

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11. 2022 **2 MARKS**

A ladder leans against a wall. The ladder makes an angle 60° with the floor. Length of the ladder is 6 metres.

(a) What is the height of the top of the ladder from the ground? **(1)**

(b) How far is the foot of the ladder from the wall? **(1)**

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12. 2022 **4 MARKS**

Two sides of a parallelogram are 20 centimetres and 10 centimetres. If the angle between them is 40°,

(a) What is the height of the parallelogram? **(2)**

(b) Find the area of the parallelogram. **(2)**

(sin 40° = 0.64, cos 40° = 0.77, tan 40° = 0.84)

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13. 2022 **1 MARK**

What is the value of tan x if x = 30°?

(1/2; 1/√2; 1/√3; √3)

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14. 2022 **1 MARK**

If the perimeter of a triangle is 24 centimetres and its inradius is 2 centimetres, find its area in square centimetres.

(12; 20; 24; 26)

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15. 2021 **1 MARK**

Measure of the smallest angle of a right angled triangle is 30°. Length of its smallest side is 6 centimetres. What is the length of its largest side?

(6, 3, 18, 12)

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16. 2021 **3 MARKS**

In triangle ABC, ∠B = 90°, AB = 7 centimetres, BC = 24 centimetres, AC = 25 centimetres.

(a) sin A = 24/K, what number is K? **(1)**

(b) Write cos C and sin C. **(2)**

17. 2021 **4 MARKS**

In triangle ABC, ∠B = 90°, BC = 1 centimetre, sin A = 1/2.

(a) What is the length of AC? **(1)**

(b) Find the length of AB. **(1)**

(c) What is the measure of ∠A? **(1)**

(d) sin 60° = _____ **(1)**

18. 2021 **5 MARKS**

In triangle ABC, AB = AC = 4 centimetres, ∠A = 120°.

(a) ∠B = _____ **(1)**

(b) Find the perpendicular distance from A to BC. **(2)**

(c) Find the area of the triangle. **(2)**

19. 2021 **5 MARKS**

A man standing at the edge of a river sees the top of a tree at an elevation of 60°. Stepping 20 metres back he sees it at an elevation of 30°. Draw a rough figure and find the width of the river.